



Dimensional Modeling: Advanced Topics





In last lecture we studied

- Requirement definition to design
- Star schema
 - Fact tables
 - Dimension tables
 - Surrogate key
 - Grain of fact table
- Advantages of star schema



Today's lecture objective:

- Some examples of star schema
- Slowly changing dimensions
- Snowflake schema
 - Options to normalize
 - Cons and pros
- Aggregating fact table
 - Multi-way aggregate fact tables
 - Effect of sparsity on aggregation
 - Goals for aggregation strategy
- Families of stars
- Snapshot and transaction table
- Core and custom tables

Product

<u>Product _Code</u>	Description	Color	Size
100	Sweater	Blue	40
110	Shoes	Brown	10 1/2
125	Gloves	Tan	M
• • •			

Period

<u>Period _Code</u>	Year	Quarter	Month
001	1999	1	4
002	1999	1	5
003	1999	1	6
• • •			

Sales

<u>Product _Code</u>	<u>Period _Code</u>	<u>Store _Code</u>	Units _Sold	Dollars _Sold	Dollars _Cost
110	002	S1	30	1500	1200
125	003	S2	50	1000	600
100	001	S1	40	1600	1000
110	002	S3	40	2000	1200
100	003	S2	30	1200	750
• • •					

Store

<u>Store _Code</u>	Store _Name	City	Telephone	Manager
S1	Jan's	San Antonio	683-192-1400	Burgess
S2	Bill's	Portland	943-681-2135	Thomas
S3	Ed's	Boulder	417-196-8037	Perry
• • •				



Slowly changing dimensions

- Type 1
- Type 2
- Type 3
 - By using star schema

Type 1 Changes

➤ General principles

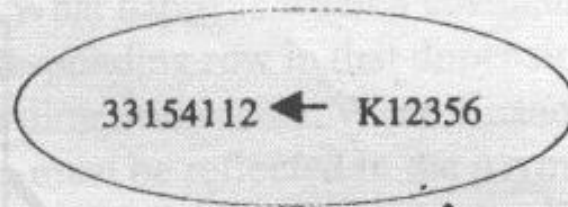
- Usually, the changes relate to correction of errors in source systems
- Sometimes the change in the source system has no significance
- The old value in the source system needs to be discarded
- The change in the source system need not be preserved in the data warehouse

➤ Applying type 1 changing

- Overwrite the attribute value in the dimension table row with the new value
- The old value of the attribute is not preserved
- No other changes are made in the dimension table row
- The key of this dimension table or any other key values are not affected
- This type is easiest to implement

KEY RESTRUCTURING

INCREMENTAL LOAD -- TYPE 1 CHANGE



Customer Code:
K12356

Customer Name:
Kristin Samuelson

BEFORE

Customer Key: 33154112

Customer Name: Kristin Daniels

Customer Code: K12356

Marital Status: Single

Address: 733 Jackie Lane,
Baldwin Harbor

State: NY

Zip: 11510

AFTER

33154112

Kristin Samuelson

K12356

Single

733 Jackie Lane,
Baldwin Harbor

NY

11510



Type 2 Changes

➤ General principles

- They usually relate to true changes in source systems
- There is a need to preserve history in the data warehouse
- This type of change partitions the history in the data warehouse
- Every change for the same attribute must be preserved

➤ Applying type 2 changes

- Add a new dimension table row with the new value of the changed attribute
- An effective date field may be included in the dimension table
- There are no changes to the original row in the dimension table
- The key of the original row is not affected
- The new row is inserted with a new surrogate key

KEY RESTRUCTURING

INCREMENTAL LOAD -- TYPE 2 CHANGES ON 10/1/2000 & 11/1/2000



Customer Code: K12356

Marital Status: Married

Address: 1417 Ninth Street,
Sacramento

State: CA Zip: 94236

BEFORE

Customer Key: 33154112
Customer Name: Kristin Daniels
Customer Code: K12356
Marital Status: Single
Address: 733 Jackie Lane,
Baldwin Harbor
State: NY
Zip: 11510

AFTER-Eff. 10/1/2000

51141234
Kristin Samuelson

K12356

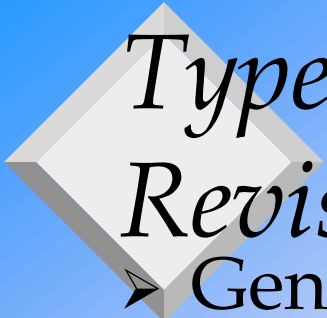
Married
733 Jackie Lane,
Baldwin Harbor
NY
11510

AFTER-Eff. 11/1/2000

52789342
Kristin Samuelson

K12356

Married
1417 Ninth Street,
Sacramento
CA
94236



Type 3 Changes : Tentative Soft Revisions

➤ General principles

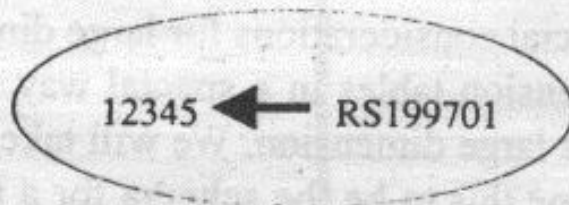
- They usually relate to "soft" or tentative changes in the source systems
- There is a need to keep track of history with old and new values of the changed attribute
- They are used to compare performances across the transition
- They provide the ability to track forward and backward

Type 3 Changes contd...

➤ Applying type 3 changes

- Add an "old" field in the dimension table for the affected attribute
- Push down the existing value of the attribute from the "current" field to the "old" field
- Keep the new value of the attribute in the "current" field
- Also, you may add a "current" effective date field for the attribute
- The key of the row is not affected
- No new dimension row is needed
- The existing queries will seamlessly switch to the "current" value
- Any queries that need to use the "old" value must be revised accordingly
- The technique works best for one "soft" change at a time
- If there is a succession of changes, more sophisticated techniques must be devised

KEY RESTRUCTURING



INCREMENTAL LOAD -- TYPE 3 CHANGE Eff. 12/1/2000

Salesperson ID:
RS199701

Territory Name:
Chicago

BEFORE

Salesperson Key 12345
Salesperson Name: Robert Smith
Old Territory Name:
Current Territory Name: New England
Effective Date: January 1, 1998
Region Name: North

AFTER

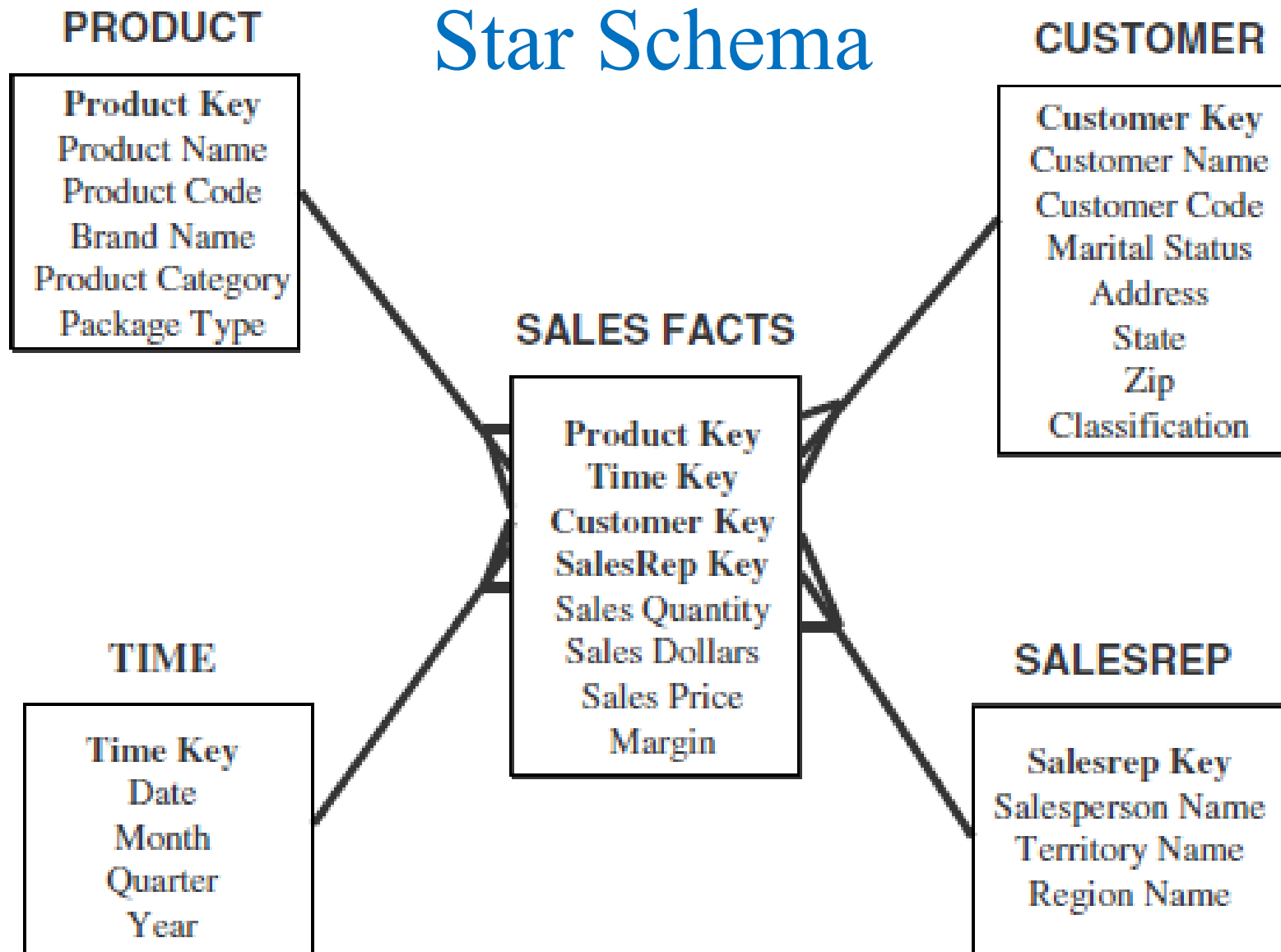
12345
Robert Smith
New England
Chicago
December 1, 2000
North

Figure 11-4 Applying Type 3 changes.

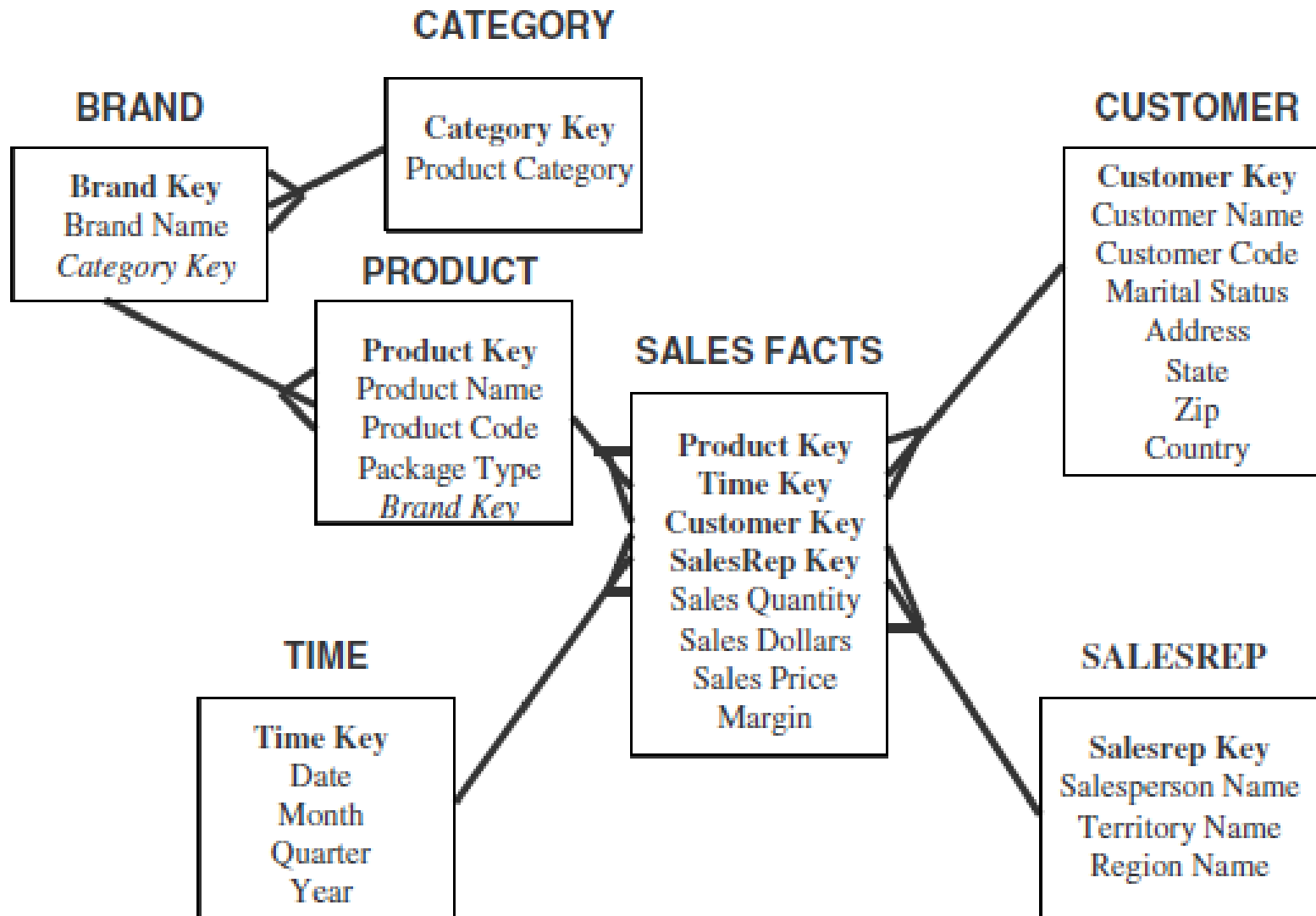


Snowflake Schema

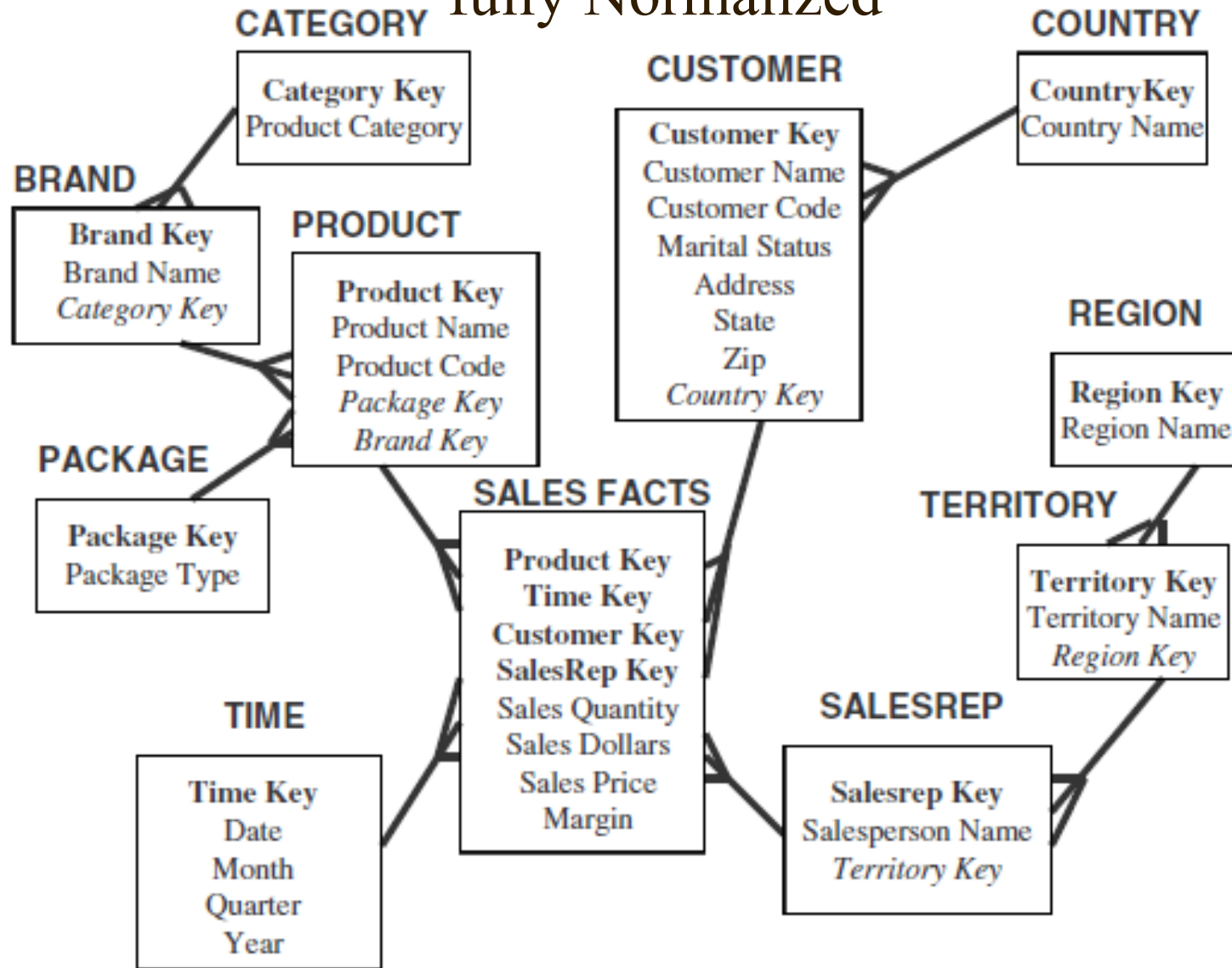
Star Schema



Snow flake Schema: Partially Normalized



Snow flake Schema: fully Normalized





Snowflake Schema contd...

- Options for making snowflake schema
 - Partially normalize only a few DTs leaving others intact
 - Partially or fully normalize only a few DTs, leaving the rest intact
 - Partially normalize every DT
 - Fully normalize every DT



Snowflake Schema contd...

➤ Advantages

- Small savings in storage space
- Easier to update and maintain

➤ Disadvantages

- Difficult to brows through
- Additional joins
- Degraded query performance



Aggregating Fact Tables

- *One-way aggregates*
- *Two-way aggregates*
- *Three-way aggregates*

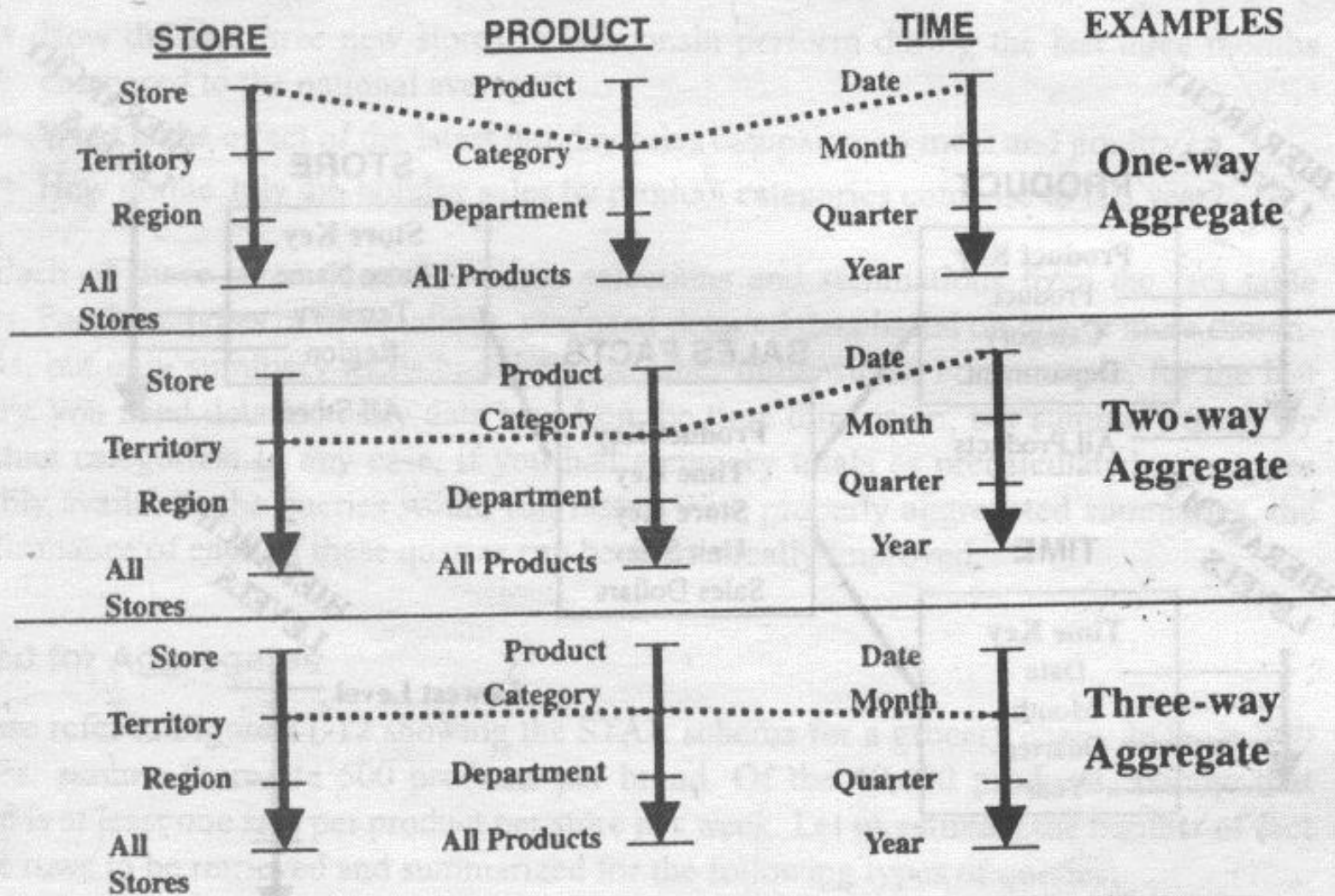


Figure 11-14 Forming aggregate fact tables.

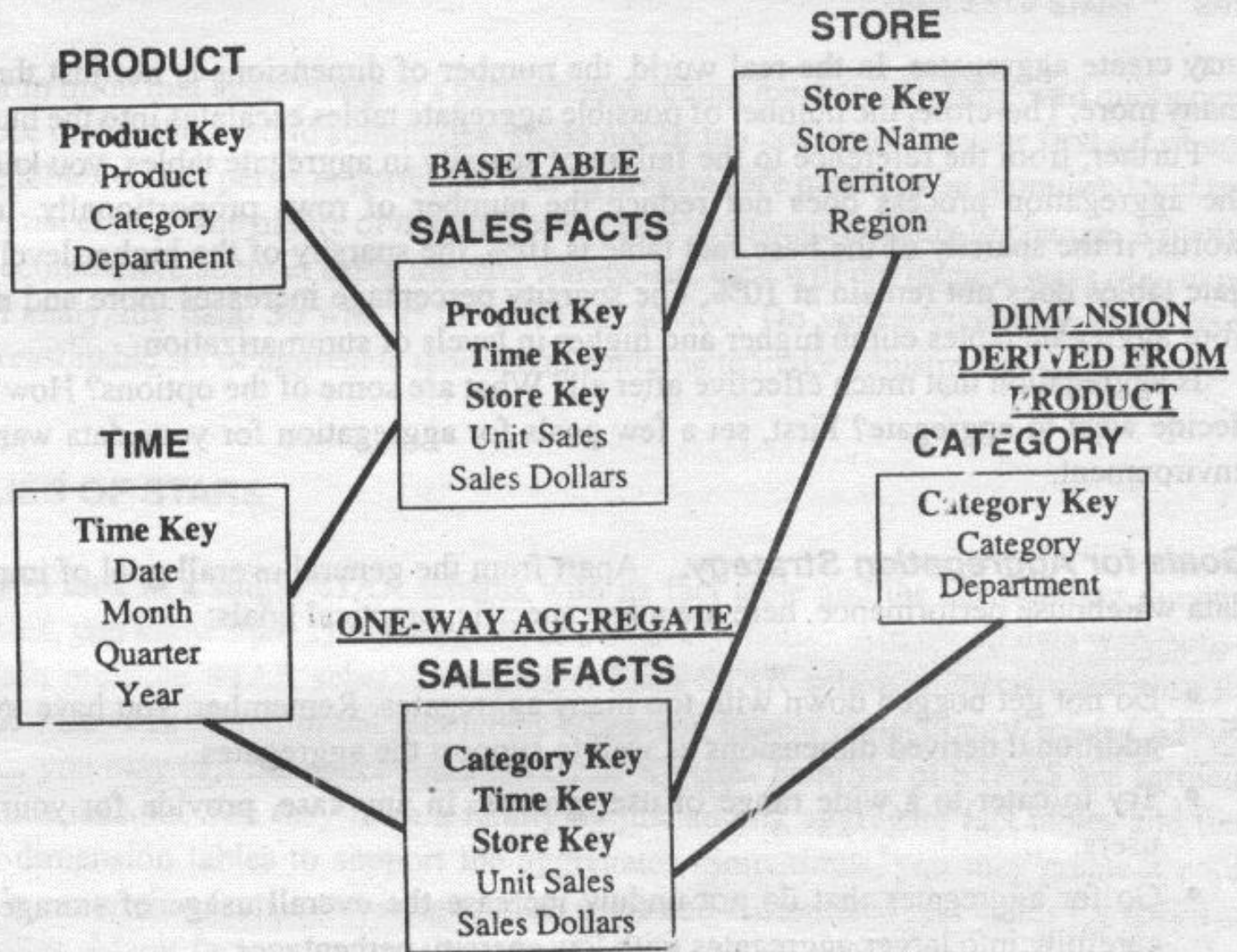
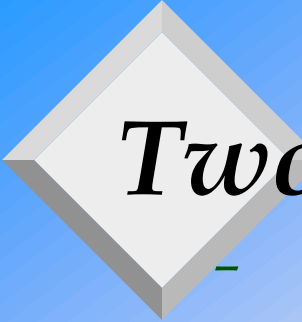


Figure 11-15 Aggregate fact table and derived dimension table.



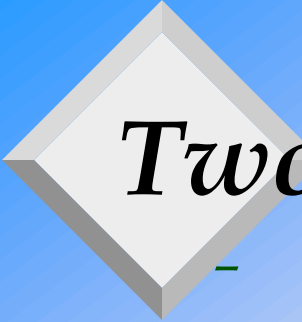
One-way aggregation

- Product category by store by date
- Product department by store by date
- All products by store by date
- Territory by product by date
- Region by product by date
- All stores by product by date
- Month by store by product
- Quarter by store by product
- Year by store by product



Two-way aggregation

- Product category by territory by date
- Product category by region by date
- Product category by all stores by date
- Product category by month by store
- Product category by quarter by store
- Product category by year by store
- Product department by territory by date
- Product department by region by date
- Product department by all stores by date
- Product department by month by store
- Product department by quarter by store
- Product department by year by store
- All products by territory by date
- All products by region by date
- All products by all stores by date



Two-way aggregation contd...

- All products by month by store
- All products by quarter by store
- All products by year by store
- District by month by product
- District by quarter by product
- District by year by product
- Territory by month by product
- Territory by quarter by product
- Territory by year by product
- Region by month by product
- Region by quarter by product
- Region by year by product
- All stores by month by product
- All stores by quarter by product
- All stores by year by product



Three-way Aggregation

- Product category by territory by month
- Product department by territory by month
- All products by territory by month
- Product category by region by month
- Product department by region by month
- All products by region by month
- Product category by all stores by month
- Product department by all stores by month
- Product category by territory by quarter
- Product department by territory by quarter
- All products by territory by quarter
- Product category by region by quarter
- Product department by region by quarter
- All products by region by quarter



Three-way Aggregation contd..

- Product category by all stores by quarter
- Product department by all stores by quarter
- Product category by territory by year
- Product department by territory by year
- All products by territory by year
- Product category by region by year
- Product department by region by year
- All products by region by year
- Product category by all stores by year
- Product department by all stores by year
- All products by all stores by year



Effect of sparsity on aggregation

- Consider the case of the grocery chain with 300 stores, 40,000 products in each store, but only 4000 selling in each store in a day. As discussed earlier, assuming that you keep records for 5 years or 1825 days, the maximum number of base fact table rows is calculated as follows:
 - **Product = 40,000**
 - **Store =300**
 - **Time=1825**
 - Maximum number of base fact table rows = 22 billion
 - Maximum number of aggregate table rows=43,800,000



Goals for Aggregation Strategy

- Do not get bogged down with too many aggregates.
- Try to cater to a wide range of user groups. In any case, provide for your power users.
- Go for aggregates that do not unduly increase the overall usage of storage. look carefully into larger aggregates with low sparsity percentages.
- Keep the aggregates hidden from the end-users. That is, the aggregates must be transparent to the end-user query. The query tool must be the one to be aware of the aggregates to direct the queries for proper access.
- Attempt to keep the impact on the data staging process as less intensive as possible



Thank You Very Much